of the mapping). The iteration may be recast in the form

$$\boldsymbol{b}_k = -(\boldsymbol{g}(z_n^{(k)}) - \boldsymbol{\tau}_n), \qquad \boldsymbol{J}^{(k)} \Delta z_k = \boldsymbol{b}_k, \qquad z_{n+1}^{(k+1)} = z_{n+1}^{(k)} + \Delta z_k.$$

The potentially demanding steps here are the calculation of the Jacobian matrix and the solution of the linear system.

One way to save computational work is to recycle the Jacobian from a previous timestep. Alternatively, it may be possible to approximate the Jacobian crudely in such a way that the iteration still converges. For example, if the Jacobian matrix is large and can be written in the form

$$\boldsymbol{J}(z) = \boldsymbol{D} + \boldsymbol{E}(z),$$

where $\boldsymbol{E}$ is small in norm and $\boldsymbol{D}$ is a constant sparse matrix, then, in many cases, the Jacobian matrix may be replaced by the constant matrix $\boldsymbol{D}$ and the iteration will still converge. Newton's method (without approximation of the Jacobian matrix) has a remarkable *quadratic* convergence property, meaning that, when the initial guess is close the solution, the errors e_k, e_{k+1} at the kth and $k+1$st iterations satisfy the relation

$$e_{k+1} \leq K e_k^2.$$

This rapid convergence is typically lost when the Jacobian matrix is approximated in some way, and one finds instead

$$e_{k+1} \leq \rho e_k,$$

where $0 < \rho < 1$, i.e., quadratic convergence is replaced by geometric convergence.

In some cases *semi-implicit* methods can be developed which may only require the solution of a low-dimensional nonlinear system at each step. In Chap. 4, we discuss constrained systems for which implicit methods are needed and, in the case of the SHAKE method, for which the nonlinear system that must be solved at each step is of dimension equal to the number of constraints imposed. This is an example of a semi-implicit method.

In other cases, the special structure of the underlying problem may lead to certain efficiencies in the implementation, as when a sparse matrix is obtained in the linear equations that must be solved at each timestep.

Other methods may be substituted for Newton's method for the purpose of solving the nonlinear equations [292] of an implicit method.